UNTITLED WORKBOOK

```
#import libraries
import pandas as pd
import matplotlib as pyplot
```

```
df = pd.read_excel('ecommerce_sales_raw.xlsx', sheet_name = 'sales_data')

print('Columns:')
print(df.columns)
print('\nShape:')
print(df.shape)
print('\nData Types:')
print(df.dtypes)
```

```
Columns:
Index(['order_id', 'customer_name', 'customer_email', 'region',
       'product_category', 'product_name', 'quantity', 'unit_price',
       'discount_pct', 'order_date', 'ship_date', 'payment_method',
       'order_status', 'sales_rep'],
      dtype='object')

Shape:
(70, 14)

Data Types:
order_id             object
customer_name        object
customer_email       object
region               object
product_category     object
product_name         object
quantity              int64
unit_price            int64
discount_pct        float64
order_date           object
ship_date            object
payment_method       object
order_status         object
sales_rep            object
dtype: object
```

DataFrames and CSVs DataFrame as df

```
SELECT *
FROM df;
```

		customer_...	customer_email	region	product_ca...	product_name
0	ORD-1001	Alice Mwangi	alice.mwangi@email.com	East Africa	Electronics	Wireless Earbuds
1	ORD-1002	BOB KARIUKI	bob.kariuki@GMAIL.COM	East Africa	electronics	Laptop Stand
2	ORD-1003	Carol Odhiambo	carol.o@yahoo.com	West Africa	Home & Kitchen	Blender
3	ORD-1004	david mwenda	david.mwenda@email.com	East Africa	Fashion	Running Shoes
4	ORD-1005	Eve Akinyi	eve.akinyi@email.com	Southern Africa	Electronics	Phone Case
5	ORD-1006	Frank Osei	frank.osei@company.com	West Africa	Beauty	Moisturizer Set
6	ORD-1007	Grace Mutua	grace.mutua@email.com	East Africa	Home & Kitchen	Coffee Maker
7	ORD-1008	Henry Oduya	null	East Africa	Electronics	Wireless Earbuds
8	ORD-1009	Irene Wanjiku	irene.w@gmail.com	East Africa	Fashion	Handbag
9	ORD-1010	James Boateng	j.boateng@email.com	West Africa	Electronics	Laptop Stand
10	ORD-1011	KAREN NDUNG'U	karen.n@email.com	East Africa	Beauty	Lip Gloss Pack
11	ORD-1012	Liam Mensah	liam.mensah@gmail.com	West Africa	Home & Kitchen	Blender
12	ORD-1013	Maria Kamau	maria.kamau@email.com	East Africa	Electronics	Bluetooth Speaker
13	ORD-1014	Nadia Owusu	nadia.owusu@email.com	West Africa	Fashion	Ankara Dress
14	ORD-1015	Oscar Kimani	oscar.kimani@email.com	East Africa	Electronics	Phone Case
15	ORD-1016	Priya Sharma	priya.s@email.com	South Asia	Beauty	Moisturizer Set

Rows: 70 Expand

DataFrames and CSVs DataFrame as df1

```
CREATE OR REPLACE TABLE sales_clean AS
--Data Cleaning--
SELECT order_id,
    LOWER(customer_name) AS customer_name,
    LOWER(COALESCE(customer_email, 'Unknown')) AS customer_email,
    LOWER(region) AS region,
    LOWER(product_category) AS product_category,
    product_name,
    quantity,
    unit_price,
    COALESCE(
        NULLIF(TRIM(CAST(discount_pct AS VARCHAR)), ''),
        '0'
    )::DOUBLE AS discount_pct,
    CAST(
        CASE
            WHEN order_date LIKE '____-__-__T%' THEN LEFT(order_date, 10)
            WHEN order_date LIKE '____-__-__'   THEN order_date
            WHEN order_date LIKE '__/__/____'    THEN strftime(strptime(order_date, '%m/%d/%Y'), '%Y-%m-%d')
            WHEN order_date LIKE '__-__-____'    THEN strftime(strptime(order_date, '%m-%d-%Y'), '%Y-%m-%d')
            ELSE                                      strftime(strptime(order_date, '%B %d, %Y'), '%Y-%m-%d')
        END AS DATE
    ) AS order_date,
    CASE
        WHEN ship_date LIKE '____-__-__%' THEN CAST(ship_date AS DATE)
        WHEN ship_date LIKE '__/__/____'   THEN strptime(ship_date, '%m/%d/%Y')::DATE
        WHEN ship_date LIKE '__-__-____'   THEN strptime(ship_date, '%m-%d-%Y')::DATE
        WHEN ship_date IS NULL             THEN NULL
        ELSE                                    strptime(ship_date, '%B %d, %Y')::DATE
    END AS ship_date,
    UPPER(order_status) AS order_status,
    LOWER(payment_method) AS payment_method,
    sales_rep
FROM df;

-- Check out the new clean sales data --
SELECT * FROM sales_clean LIMIT 5;

--analysis--
-- which region generates the most revenue --
SELECT region,
    SUM(quantity * unit_price * (1 - discount_pct/100)) AS total_revenue
FROM sales_clean
    WHERE region IS NOT NULL
    GROUP BY region
    ORDER BY total_revenue DESC; -- west africa leads with a total_revenue of 447695

-- which product category has the highest average order value --
SELECT product_category,
    COUNT(quantity) AS quantity_per_category,
    AVG(quantity * unit_price * (1 - discount_pct/100)) AS avg_revenue
FROM sales_clean
WHERE product_category IS NOT NULL
AND order_id != 'ORD-1061'
GROUP BY product_category
ORDER BY avg_revenue DESC;

-- total revenue per category --
SELECT product_category,
    COUNT(order_id) AS total_orders,
    ROUND(SUM(quantity * unit_price * (1 - discount_pct/100)), 2) AS total_revenue
FROM sales_clean
WHERE product_category IS NOT NULL
AND order_id != 'ORD-1061'
GROUP BY product_category
ORDER BY total_revenue DESC;

-- Which sales rep closed the most orders and generated the most revenue --
SELECT sales_rep,
    COUNT(order_id) AS total_orders,
```

```sql
    ROUND(SUM(quantity * unit_price * (1 - discount_pct/100)), 2) AS total_revenue
FROM sales_clean
WHERE product_category IS NOT NULL
AND order_id != 'ORD-1061'
GROUP BY sales_rep
ORDER BY total_revenue DESC;

-- What is the order fulfilment breakdown?
-- How many orders were Delivered, Cancelled, and Processing --
SELECT order_status,
    COUNT(*)
FROM sales_clean
GROUP BY order_status
    ORDER BY order_status DESC;

-- top 5 customers by total spend --
SELECT customer_name,
ROUND(SUM(quantity * unit_price * (1 - discount_pct/100)), 2) AS total_spend
FROM sales_clean
    WHERE order_id != 'ORD-1061'
    GROUP BY customer_name
ORDER BY total_spend DESC
    LIMIT 5;

-- average number of days between order date and ship date --
SELECT
    ROUND(AVG(DATEDIFF('day', order_date, ship_date)), 1) AS avg_days_to_ship
FROM sales_clean
WHERE ship_date IS NOT NULL
AND order_id != 'ORD-1061';

--month with the highest sales volume
SELECT
    STRFTIME(order_date, '%B') AS month,
    MONTH(order_date) AS month_num,
    COUNT(order_id) AS total_orders,
    ROUND(SUM(quantity * unit_price * (1 - discount_pct/100)), 2) AS total_revenue
FROM sales_clean
WHERE order_id != 'ORD-1061'
GROUP BY month, month_num
ORDER BY month_num ASC;

-- orders that are still in Processing status with no ship date --
SELECT order_id,
    customer_name,
    STRFTIME(order_date, '%Y-%m-%d') AS order_date,
    order_status
FROM sales_clean
WHERE order_status = 'PROCESSING'
AND ship_date IS NULL;
```

...	...	custo...	o...	orde...
0	ORD-1006	frank osei	2024-01-18	PROCESSING
1	ORD-1026	zara ndegwa	2024-03-12	PROCESSING
2	ORD-1037	kemi williams	2024-04-10	PROCESSING
3	ORD-1059	felix osei	2024-06-07	PROCESSING

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